Working Groups (all)

GY457 Autumn Term 2025 (Seminar Topic 1)

Lecturer: *Dr Sara Bagagli*

Context[1]

This seminar builds on the lecture on Topic 1: "Regional and urban concentration forces". During the lecture, we have discussed a range of theories to explain why firms and residents concentrate spatially. Agglomeration forces, by raising productivity, make dense locations more attractive and increase demand for space. The purpose of this seminar is to discuss how shifts in the demand for space across space and time impact on prices and quantities on real estate markets.

"Homework" to be carried out prior to the seminar

The compulsory reading for this seminar is:

Chapter 1 in DiPasquale D., and W. Wheaton, 1996, *Urban Economics and Real Estate Markets*, Prentice Hall, Englewood Cliffs (N.J.). Available from LSE Reading List.

All students are expected to self-study the book chapter and familiarise themselves with the four-quadrant diagram and the underlying theoretical assumptions, mechanisms, and predictions. With the help of the diagram, all students are expected to engage with the seminar tasks below. All students must be prepared to answer the questions during the seminar. It is recommended that they prepare some notes. However, no submission of written answers is expected. *Presenters*, in addition, are expected to prepare slides for a joint presentation of their answers (one presentation per seminar group). In doing so, they may split the tasks or work together on all tasks as they prefer (the latter is the recommended option).

Seminar tasks:

Necessarily all students including the presenters are requested to answer the below questions using the graphical version of the model. It is recommended that, in addition, students answer the questions mathematically with reference to the scenario attached.

1) Draw a variant of the four-quadrant diagram for an exemplary real estate market (let's think of the market as referring to a metropolitan area). Illustrate the equilibrium outcomes in each of the four quadrants. Please pay attention to

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appropriate labelling of axes and curves. *For presenters only: Discuss how each of the curves in the four diagrams is derived from theoretical assumptions.*

2) Now compare the equilibrium in the baseline case from 1) to a scenario in which the local planner is more restrictive, i.e. it is more difficult to obtain planning permission (to develop). How do the equilibrium outcomes change?
3) Now assume that there is a shift in the national economy towards the production of knowledge-based tradable services. Use the four-quadrant model to derive the impact on the local real estate market (the four equilibrium outcomes) in two scenarios:
    a. A large and dense city that offers good access to "knowledge" (high-skilled) workers and (innovative) firms
    b. A smaller city largely specialized on manufacturing standardized mass-produced goods
4) Focusing on the large "knowledge" city, now distinguish the effects of the exogenous macro-economic shock (the shift towards knowledge-based tradable services) in two scenarios where the planner is either more or less restrictive (it is more or less difficult to obtain planning permission).
5) Against the macro-economic trend (shift towards "interactive" knowledge-based tradable services), in what type of cities (big-restrictive, big-less restrictive, small-restrictive, small-less restrictive) should you concentrate your economic activities if you were
    a. A real estate investor wishing to maximize the returns viewing real estate as an investment asset?
    b. A construction company making money from being contracted to construct office and residential buildings?
    c. A developer buying and developing land selling residential and office space?

During the seminar

After an introduction, the presenters will walk the group through their answers during an "interactive" presentation. During the presentation, it is expected that the presenters "engage" with the audience, i.e. prompt the audience to suggest answers. Likewise, the audience is expected to challenge the answers provided by the presenters. *The presenters are kindly asked to make use of interactive elements such as Menti to engage the group.* Before they give their own answers, they may choose to ask the audience to answer the question in a poll to test the understanding. Other "interactive" elements are encouraged.

All presenters should carry an approximately similar weight in the presentation. In preparing the presentation, presenters may choose to split the tasks or work as a group. The lecturer will monitor the presentation and the discussion and intervene if appropriate. The presenters are expected to make notes during the discussion in case corrections to their suggested answers are due. Given that the presentation is supposed to be interactive, a net-presentation time of approximately 60 minutes is advised.

After the presentation, we may move on to a more general discussion of the determinants of demand and supply on real estate markets and how urban economics may help with better explaining real estate market outcomes.

After the seminar

If indicated by the feedback during the seminar, presenters will revise their presentation. Within 24 hours, they will then ask the lecturer to upload the slides onto the GY457 Moodle page. Please, use the following filenames to save your presentations: Topic_1_group_X, where X indicates the seminar group.

Scenario

Consider an urban economy described by the following system of log-linearized equations.

(1) $$\ln(D) = \ln a - b\ln(R)$$

(2) $$\ln(P) = \ln(R) - \ln(i)$$

(3) $$\ln(C) = \ln d + e\ln(P)$$

(4) $$\ln(S) = \ln(C) - \ln(\delta)$$

The following parameter values apply:

a = 5,000,000,000 (m^2); b = 0.5; i = 0.05; d = 2,000 (£/(m^2)); e = 1; δ = 0.025

For the vector of variables $X = \{D, R, P, C, S\}$ the notation used in DiPasquale and Wheaton (1996) applies.

Adding to Q1: Solve the system of equation for the equilibrium values of X^*. Note: A correct solution will equate demand to supply on the real estate market.

Adding to Q2: In answering the question, assume that the elasticity of construction with respect price is 10% lower than in the baseline scenario

Adding to Q3: Assume that wages in knowledge-based tradable services (prime services) will grow at a nationally uniform rate of 20%, whereas manufacturing wages with grow at a rate of uniform rate of -20%. The share of prime services is 80% in the big city and 20% in the small city. Workers are immobile (there is migration) and labour markets clear (there is no unemployment).

Adding to Q4: Combine the scenario growth of the services city from Q3 with the price elasticity of construction from Q2.